

claude-windows / 6a82308f-69d6-43a8-9ea4-d8eaf497253d

Metadata

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- Source: `claude-windows`  
- Kind: `claude`  
- Source file: `C:\Users\avidu\.claude\projects\C--Users-avidu-Projects-badminton-highlight-indexer\6a82308f-69d6-43a8-9ea4-d8eaf497253d\subagents\workflows\wf_736be611-fb5\agent-ace4bc500f0cafbc8.jsonl`  
- SHA-256: `a87318d57981fc6e6f8d2b8a19cefcd56b63412fabd23116f092f2c58d67532`  
- Source modified: `2026-06-16T15:13:57+00:00`  
- Imported at: `2026-07-05T16:48:27+00:00`  
- project: `wf_736be611-fb5`  
- session_id: `6a82308f-69d6-43a8-9ea4-d8eaf497253d`
```

Transcript

1. user (2026-06-16T15:12:42.306Z)

You are a temporal-action-localization / segmentation expert. Analyze the WINDOWING behavior of the local rally detector on real footage: the over-merge -> over-segment pendulum (baseline under-segments to 48 mega-windows; W1 over-corrects to 250 vs Gemini 165), the strong PER-VENUE pattern (GX court over-splits ~2x, mahadevpura-2 lands ~right, mahadevpura-1 stays a blob), W2's count-vs-recall tradeoff, and the hard-cap. What does this say about the cap=12 choice, the velocity/merge-gap knobs, and whether a single global windowing can fit all venues? Cite the numbers.

REAL-FOOTAGE VALIDATION DATA (6 Kushagra/Yash June-12 clips, 1080p proxies, fully-LOCAL no-AI WASB windowing vs Gemini. Gemini = strong REFERENCE, NOT ground truth ? these 6 have no golden labels EXCEPT mahadevpura-2 which the owner hand-labeled, see GROUND-TRUTH below).

Rally COUNT (Gemini | local baseline | local W1 cap=12 | local W1+W2 mc=1):
GX010128: 39 | 15 | 78 | 75
GX010129: 27 | 2 | 50 | 47
GX020125: 11 | 6 | 18 | 15
mahadevpura-0: 40 | 19 | 62 | 43
mahadevpura-1: 9 | 1 | 2 | 2
mahadevpura-2: 39 | 5 | 40 | 40
AGGREGATE: Gemini 165 | baseline 48 (mean window 62.9s, 27 merge-groups, mIoU 0.23) |
W1 250 (mean 10.8s, 10 merge-groups, mIoU 0.39) |
W1+W2 222 (mean 11.5s, 10 merge-groups, mIoU 0.39, 6 gemini-only "misses")
Per-video mean window (W1): GX010128 7.8s, GX010129 9.3s, GX020125 14.1s, mahadevpura-0 10.4s,
mahadevpura-1 86.9s (the continuously-active blob W1 cannot cap ? no inactivity gap),
mahadevpura-2 13.8s. Gemini means ~5.7-7.2s.

GROUND-TRUTH (mahadevpura-2, 40 owner-labeled rallies, mean 5.7s, verified frame-aligned):
F1@0.5 / F1@0.25 / mean-IoU: Gemini 0.557/0.835/0.55 ; local W1+W2 0.150/0.500/0.31 ;
local W1+W2+hard-cap 0.296/0.722/0.50 (best local). baseline 0.000/0.000/0.03.
BOUNDARY ERROR (matched rallies, seconds): Gemini |?start| med 1.46 |?end| med 1.19 ;
local-hardcap |?start| med 1.74 (~= Gemini!) |?end| med 3.12 (>> Gemini).
=> local's rally-START detection is ~at par with Gemini; the gap is the rally-END.
IoU<->time: for a 5.7s rally, tIoU 0.5 tolerates only ~+-1.9s shift (the ~1-1.5s service window
nearly exhausts it -> tIoU 0.5 over-penalizes; owner wants a +-2s tolerance metric).

PRIOR GOLDEN DELTAS (n=6, then n=7 incl. mahadevpura-2): baseline F1 0.300 -> W1 0.439 (+0.139, 6/0)
-> W1+W2 0.457 (+0.019, 6/0); hard-cap golden-neutral on n=6 (+0.006 n.s.) but became BEST on n=7
(continuously-active clips). All flags default-OFF, promoted only on measured LOVO lift +

sign-test.

WINDOWING KNOBS: a frame is "active" iff shuttle visible AND per-second-normalized velocity > velocity_thresh(0.02); active frames within merge_gap(2.0s) coalesce; min_window(2.0s). W1 = max_window_duration cap(12s) split at the largest internal inactivity gap. W1.5 hard-cap = midpoint-chop when no gap. W2 = net-crossings gate (drop windows with <1 net crossing).

For raw per-video detail you MAY Read C:/Users/avidu/Projects/badminton-highlight-indexer/output/local_vs_gemini_6.json . Treat the DATA block as authoritative for the numbers. Constraints: local pipeline must stay MIT/Apache/BSD-only, single local GPU, ~6-7 golden videos now (~16 by the weekend), Gemini usable at inference only (never as a training-data source).

2. assistant (2026-06-16T15:12:51.885Z)

I'll analyze the windowing data. Let me first read the raw per-video detail to ground my analysis in the actual numbers.

3. user (2026-06-16T15:12:52.350Z)

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```

4. assistant (2026-06-16T15:13:01.285Z)

I have what I need. The DATA block plus the raw JSON give me per-video and per-variant detail. Let me synthesize the windowing analysis.

Key derived numbers I'll lean on:

- The pendulum: baseline 48 (mean window 62.9s, 27 merge-groups, mIoU 0.23) ? W1 250 (mean 10.8s, mIoU 0.39) ? W1+W2 222 vs Gemini 165.
- Per-venue split: GX clips over-split ~2x (GX010128 78/39 = 2.0x, GX010129 50/27 = 1.85x, GX020125 18/11 = 1.64x); mahadevpura-2 lands ~right (40/39 = 1.03x); mahadevpura-0 1.55x?1.075x after W2; mahadevpura-1 the blob (2 windows, mean 86.9s, 0 matched, 9 gem rallies in merges).
- The cap=12 vs the per-video means; the END-boundary error (?end med 3.12s >> Gemini 1.19s); the mahadevpura-1 no-gap failure.

5. user (2026-06-16T15:13:44.950Z)

Structured output provided successfully

6. assistant (2026-06-16T15:13:57.192Z)

I've completed the windowing analysis and returned it via StructuredOutput. Summary of what the numbers say:

The over-merge ? over-segment pendulum is real and global, swinging from baseline's 0.29x of Gemini (mega-windows, mIoU 0.23) straight past 1.0x to W1's 1.52x ? never landing near parity in aggregate. But the over-segmentation is **per-venue**: the 3 GoPro (GX) clips all over-split 1.64?2.00x under W1, while mahadevpura clips span 0.22x (the blob) to 1.55x, with mahadevpura-2 landing at 1.03x.

cap=12 is the proximate problem: true/Gemini rallies are ~5.7?7.2s (owner GT 5.7s), so a 12s cap is ~2x too coarse ? it carries dead time onto rally ends (|?end| med 3.12s vs Gemini's 1.19s) while rally **starts** are already at Gemini parity (1.74s ? 1.46s). The hard-cap midpoint-chop, not the 12s gap-split, is what lifts mahadevpura-2 F1@0.5 from 0.150 to 0.296, confirming 12s is mis-sized.

W2 trades count for recall and is venue-localized: nearly its entire effect is on mahadevpura-0 (62?43, +5 gemini-only misses); it's a 3-window trim on each GX clip and fully inert on the blob. mahadevpura-1 proves the structural ceiling ? continuously active, no inactivity gap, so the cap never fires (mean window 86.9s, 0 matched, mIoU 0.0).

The verdict on a single global windowing: it can't fit all venues. The same knobs that under-split the blob (0.22x) over-split GX (2.0x), with a mean-window spread of 7.8s?86.9s under one

config ? that spread is the signature of one global setting failing across venues. The full findings, evidence, and recommendations (resize cap toward 6?8s, attack the END boundary specifically, gate on F1/matched not count parity, adopt a $\pm 2s$ tolerance metric, build a gap-free splitter for the blob, and move to venue presets / content-adaptive caps) are in the structured output.